GSPS: Geometric Semantic Positioning System

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Abstract

I present the Geometric Semantic Positioning System (GSPS), a framework that reveals semantic structure by projecting high-dimensional embeddings onto constrained 2D geometric templates. The GSPS system uses relative positioning to place inner nodes according to relative semantic-similarity to fixed anchor nodes.

The protocol enables deterministic, framework-agnostic semantic positioning where any system with the same anchor configuration will compute identical coordinates for a given concept.

1 Core Principles

1.1 Anchor-Defined Spaces

Semantic space is bounded by 1–4 fixed anchor nodes positioned at predetermined coordinates:

- **4-Anchor:** Square configuration $[(-1,0),(1,0),(0,1),(0,-1)]$
- **3-Anchor:** Triangle configuration $[(-1,0),(1,0),(0,1)]$
- **2-Anchor:** Edge configuration $[(-1,0),(1,0)]$
- **1-Anchor:** Point configuration $[(0,0)]$

1.2 Force-Based Positioning

Inner nodes (non-anchors) are positioned through:

- Gravitational attraction when cosine similarity to an anchor exceeds the dynamic pivot
- Electromagnetic repulsion when cosine similarity falls below the dynamic pivot

Dynamic pivot is calculated as mean cosine similarity across anchors, for manifold-dependent relative positioning.

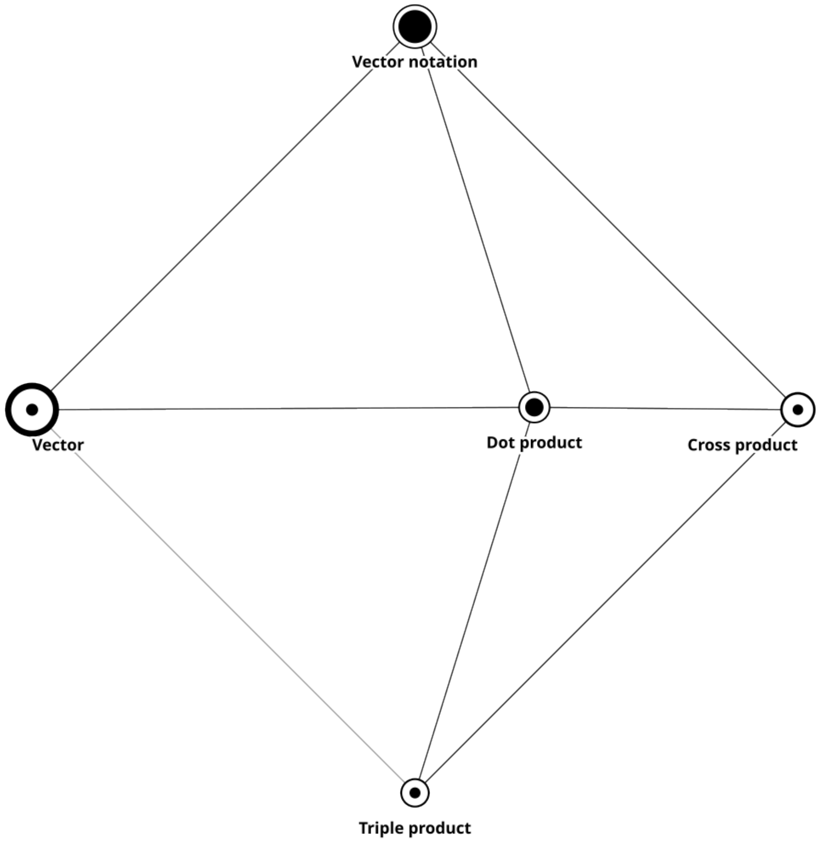
2 Topological Glyphs

To convey network connectivity alongside spatial semantics, GSPS employs a nested glyph visual encoding for each node. This bi-layered structural composition provides immediate visual context regarding a node's topological role within the graph:

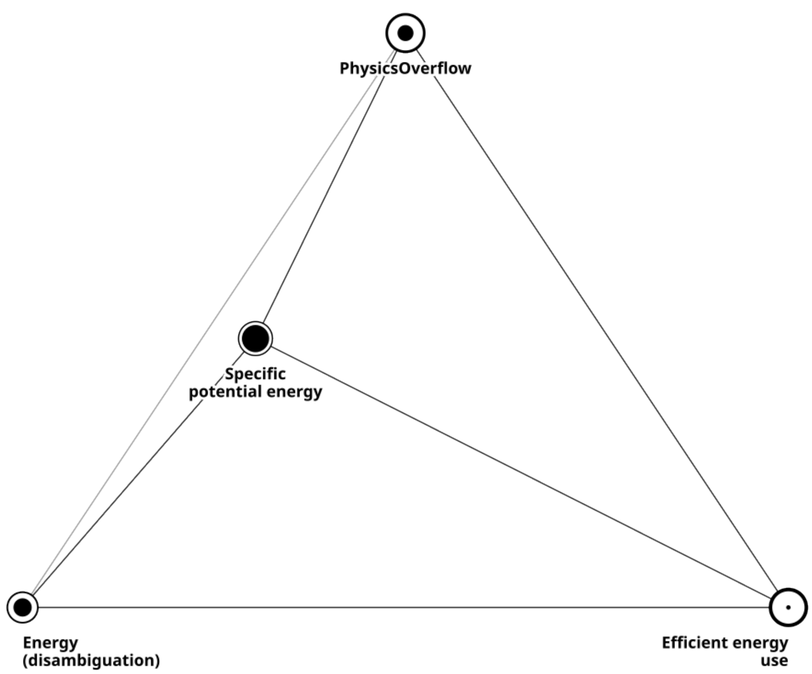
- **Outer Glyph (Out-Degree):** The overall radius and boundary weight of the node represent its outward connectivity and total structural importance.
- **Inner Glyph (In-Degree):** A solid inner core is rendered when a node acts as a destination (in-degree > 0), representing its gravitational pull as a target concept.

This dual-indicator approach allows observers to rapidly distinguish between source concepts, terminal concepts, and central hubs without relying strictly on explicit edge rendering.

3 Visual Examples

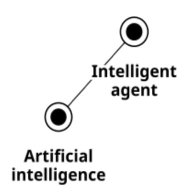


4-Anchor Square



3-Anchor Triangle

2-Anchor Edge



1-Anchor Point